

732A96/TDDE15 ADVANCED MACHINE LEARNING

EXAM 2026-08-27

TEACHER

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GRADES

- For 732A96 (A-E means pass):
 - A=19-20 points
 - B=17-18 points
 - C=14-16 points
 - D=12-13 points
 - E=10-11 points
 - F=0-9 points
- For TDDE15 (3-5 means pass):
 - 5=18-20 points
 - 4=14-17 points
 - 3=10-13 points
 - U=0-9 points

In each question, full points requires clear and well motivated answers and commented code.

INSTRUCTIONS

- This is an individual exam. No help from others is allowed. No communication with others is allowed. Answers to the exam questions may be sent to Urkund.
- This is an anonymous exam. Do not write your name on it.
- The answers to the exam should be submitted in a single PDF file. You can make a PDF from LibreOffice (similar to Microsoft Word). You can also use Markdown from RStudio (no support is provided though). Include important code needed to grade the exam (inline or at the end of the PDF file).

ALLOWED HELP

Everything on the course web page. Your individual and group solutions to the labs. This help is available on the corresponding directories of the exam system.

1. PROBABILISTIC GRAPHICAL MODELS (7 P)

In this exercise, no implementation is required. In the course, we have seen how to read the independencies represented by a DAG with the help of the d-separation criterion. An alternative is the moralization criterion. To check if a DAG G represents that X is independent of Y given Z , first, identify the ancestral set S of all nodes in $X \cup Y \cup Z$, i.e., all the nodes in $X \cup Y \cup Z$ plus all the nodes that have a directed path to a node in X , Y or Z . Second, restrict the original DAG G to the subgraph containing only the nodes in S and their corresponding edges, which we denote as G_S . Third, construct the moral graph G_S^m of G_S by adding an edge between any two non-adjacent nodes in G_S that are both parents of the same child node (sometimes described as "marrying unmarried parents"), and then dropping all edge directionalities. Finally, if all paths between X and Y in G_S^m are blocked by Z , then the original DAG G represents that X is independent of Y given Z .

Sketch a formal proof that moralization and d-separation are equivalent.

2. HIDDEN MARKOV MODELS (3+2+1 P)

Consider a hidden Markov model used to model a DNA sequence. Each position in the sequence belongs to one of two hidden states:

$$S_t \in \{\text{Background, CG-rich}\}.$$

At each position, one nucleotide is observed:

$$O_t \in \{A, C, G, T\}.$$

The initial state probabilities are

$$\pi = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}.$$

The transition matrix is

$$A = \begin{pmatrix} 0.85 & 0.15 \\ 0.25 & 0.75 \end{pmatrix},$$

where the rows represent the current state and the columns the next state.

The emission probabilities are

$$B = \begin{pmatrix} 0.30 & 0.20 & 0.20 & 0.30 \\ 0.10 & 0.40 & 0.40 & 0.10 \end{pmatrix},$$

where the columns correspond to A, C, G, T , respectively.

The observed DNA sequence is

$$O = (A, C, G, C, G, T, C, G, A, C, G).$$

Use the HMM package to construct the hidden Markov model given initial, transition, and emission probabilities. Calculate the probability

$$P(O_1, \dots, O_{11})$$

of observing the observed DNA sequence using the forward algorithm. Finally, discuss whether you would expect the effect of the initial probabilities to become smaller or larger for a very long observation sequence.

3. GAUSSIAN PROCESSES (5+1+1 P)

The derivative or gradient of a function g at a point x is defined as the limit of

$$\frac{g(x+h) - g(x)}{h}$$

when h goes to 0. It can be approximated by

$$\frac{g(x+\epsilon) - g(x-\epsilon)}{2\epsilon}$$

when ϵ is small. Use this approximation to implement gradient descent to select the bandwidth l of the squared exponential kernel of a Gaussian process for classification. You may prefer to define

$$l = \exp(K)$$

and optimize K .

Use the kernlab package and the iris dataset. Use only two classes from the iris dataset. Use the negative log-likelihood as loss function: For $y_i \in \{0, 1\}$ and predicted probability $p_i = P(y_i = 1 | x_i)$, define the loss as

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)].$$

Your code should include three functions: Loss calculation, gradient calculation, and parameter updating. You are free to make any choices that are not specified above. You do not need to run your code, but make sure that it includes appropriate comments for readability.

Finally, answer the following two questions. Do you think that the 0/1 loss function would have been a better or worse alternative as loss function for this exercise? Why is it preferable to optimize $K = \log(l)$ rather than l directly?